Online Retail Sales Analysis

SWYNEX Technologies | Final Data Analytics Project

Tools : Python, NumPy, pandas, power BI, matplotlib

Dataset : Online retail transactions.

1. Problem Statement

Retail teams need a clear way to understand revenue, order activity, product performance, and geographic sales. This project analyses historical online retail transactions and presents findings in an interactive dashboard.

2. Objectives

- Clean transaction records for analysis.
- Track revenue, orders, customers, quality, and product count.
- Explore monthly trends, top products and countries.
- Build a Power BI dashboard and communicate business insights.

3. Dataset Information

The original working dataset contained 541,909 rows and 8 columns.

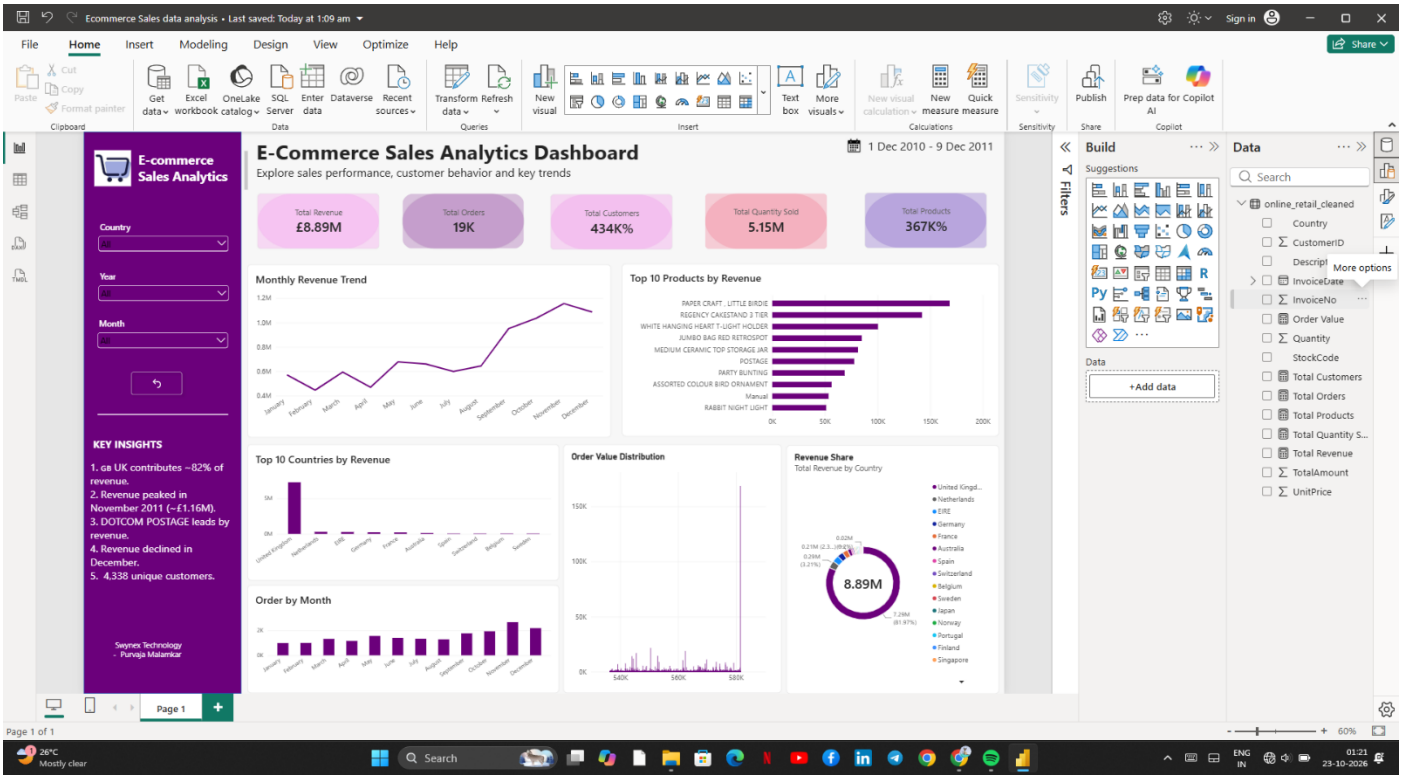
Field	Description
Invoice No	Invoice identifier
Stock Code	Product code
Description	Product description
Quantity	Units in transaction line
Invoice Date	Date and time
Unit Price	Price per unit
Customer ID	Customer identifier
Country	Customer country

4. Cleaning Process

- Checked missing values, duplicates, data types, and inconsistent values.
- Removed exact duplicate rows and trimmed text whitespace.
- Removed rows missing Description or Customer ID.
- Filtered out non-positive Quantity and Unit Price records.
- Converted Invoice Date and numeric columns to suitable types.
- Create total amount = Quantity * Unit Price.
- Exported online_retail_cleaned.csv.

5. Analysis and Dashboard

Analysis measures include revenue (sum of Total Amount), orders (distinct Invoice No), customers (distinct Customer ID), quantity sold, and distinct products. Visuals include KPI cards, Monthly Revenue Trend, Top 10 Products by Revenue, Top 10 Countries by Revenue, and Country/Year/Month slicers.



6. Key Business Insights

Earlier finding (verify in final dashboard)	Possible business action
UK contributed about 82% of revenue (estimate).	Protect the core market; assess expansion opportunities.
Revenue peaked in November 2011 (estimate ~£1.16M).	Plan inventory and operations around seasonal demand.
DOTCOM POSTAGE appeared as a high-revenue item.	Check whether it is a charge, not a merchandise product.
December revenue was lower than November in the earlier vie	Review seasonality, available trading days, nd returns.
Earlier cleaned profile: ~4,338 unique customers.	Explore repeat purchase and retention patterns.

The values above are approximate previous results. Recheck against the final dashboard and its slicer selections.

7. Recommendations

- Compare order count with revenue to understand basket-size changes.
- Validate charge-like products before interpreting product rankings.
- Prepare for seasonal demand peaks.

8. Limitations

This is historical descriptive analysis and does not prove causes. Results depend on the cleaning rules and definitions used. Customer IDs may be absent from some transactions, and filtering can affect totals.

9. Submission Checklist

- Include analysis notebook/script.
- Include the Power BI PBIX.
- Add Dashboard Screenshot.
- Include the cleaned data only if redistribution is allowed
- Upload the repository and submit its URL.

GitHub repository URL: https://github.com/purvaja0321/SWYNEX-Data-Analysis.git